FIRST TECH CHALLENGE | TEAM #18438

WOLFPACK MACHINING · ENGINEERING PORTFOLIO

Season: CenterStage (2023-24) | Location: Cary, North Carolina, USA
Organization / School: Green Level High School

MAJOR AWARDS & JUDGING RECOGNITION

· Think Award Winner · FIRST World Championship (Houston, TX)
· Inspire Award 2nd Place · World Championship
· Design Award Winner · North Carolina State Championship

EXECUTIVE SUMMARY

World Think Award winners FTC #18438 Wolfpack Machining demonstrate exceptional engineering discipline, mathematical modeling, FEA topology optimization, and master-level notebook documentation.

CORE SUBSYSTEMS & MECHANISMS

· Custom CNC Differential Bevel Wrist
· Carbon Fiber Sandwich Plate Chassis
· Magnetic Latch Intake Shroud
· Active Tensioned Constant-Force Cascade Lift

MECHANICAL & CAD ENGINEERING HIGHLIGHTS

· Custom Differential Bevel Wrist: CNC-machined bevel gearbox delivering two degrees of freedom with sub-0.2 degree backlash.
· FEA Topology Optimization: Generative design on suspension brackets and elevator mounts shaving 140g without losing structural rigidity.
· FMEA Failure Mode Analysis: Exhaustive failure logging and risk matrices applied to every subsystem before tournament deployment.

AUTONOMOUS NAVIGATION & SOFTWARE

· Cubic Spline Trajectory Generation: Native Java spline solver generating dynamic collision-free navigation paths in real time.
· Multi-Sensor Kalman Filtering: Encoders fused with 6-axis IMU to eliminate drift over high-speed match cycles.
· Telemetry Pit Dashboards: Real-time telemetry streaming motor thermal health and battery sag over custom WebSocket connections.

COMMUNITY OUTREACH & STEM ADVOCACY

CONNECTING & MOTIVATING INITIATIVES

· Open-Source FTC CAD Library: Published 35+ verified parametric assemblies on Onshape, used by more than 300 teams globally.

· International Technical Webinars: Hosted bi-weekly workshops covering custom gear calculations, FEA analysis, and autonomous control theory.

· Diversity in Engineering Summer Camps: Targeted hands-on STEM programs introducing underrepresented middle schoolers to CAD modeling.